

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279708

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Ubelhart; Eric et al.

HYBRID DRIVE RESOLVER ROTOR MOUNTING ARRANGEMENT

Abstract

A hybrid drive arrangement for an internal combustion engine is provided, and includes an emotor having a rotor that is connected to the crankshaft and a stator that is fixed relative to the engine. A flex plate is connected for rotation with the rotor, and a resolver drive shaft extends from the flex plate, coaxial with the rotor. A resolver is provided having a resolver rotor that is connected with the resolver drive shaft at a position spaced apart from the flex plate, and a resolver stator is fixed relative to the emotor stator and located in proximity to the resolver rotor. A resolver rotor support acts to support the resolver drive shaft in proximity to the resolver stator. This arrangement isolates deflections caused by the crankshaft so that the radial deflections of the resolver rotor are minimized in order to ensure proper functioning of the resolver.

Inventors: Ubelhart; Eric (Orville, OH), Rentfrow; Peter (Doylestown, OH)

Applicant: Schaeffler Technologies AG & Co. KG (Herzogenaurach, DE)

Family ID: 96880620

Assignee: Schaeffler Technologies AG & Co. KG (Herzogenaurach, DE)

Appl. No.: 18/594085

Filed: March 04, 2024

Publication Classification

Int. Cl.: H02K24/00 (20060101); B60K6/40 (20071001)

U.S. Cl.:

CPC H02K24/00 (20130101); B60K6/40 (20130101);

Background/Summary

FIELD OF INVENTION

[0001] The disclosure relates to a hybrid drive architecture for hybrid electric vehicles. More specifically, it relates to P0 and P1 hybrid drive architectures and mounting of sensors used for detecting rotational position and speed of cantilever mounted electric motor rotors.

BACKGROUND

[0002] In a P1 hybrid architecture, an electric motor (emotor) is integrated into the drivetrain with the rotor directly engaged with the crankshaft of the internal combustion engine (ICE), and torque from the emotor and/or the ICE is transmitted via a transmission assembly that includes the torque converter to the transmission gear box. In a P0 hybrid architecture, the emotor is connected to the front of the ICE

[0003] In certain P0 and P1 hybrid architectures, the emotor rotor has radial/angular movement due to the emotor rotor being mounted to the engine crankshaft in a cantilever arrangement, for example due to vibration and piston-firing induced crankshaft flexing due to the direct mounting of the emotor rotor to the crankshaft. For monitoring and control of the emotor, a resolver is mounted to the rotor in order to determine an angular velocity and position of the emotor rotor. However, if the resolver rotor is mounted too far from the base mounting surface of the rotor, it can see high radial displacement. This can cause issues with the Transformation ratio (input voltage to output voltage) due to the changing air gap between the resolver rotor and its own pick-up/stator. This can lead to signal processing issues and speed read out issues in the inverter controlling the emotor.

[0004] It would be desirable to find a cost-effective solution to these issues that improves quality and performance.

SUMMARY

[0005] In one aspect, a hybrid drive arrangement for an internal combustion engine having a crankshaft is provided. The hybrid drive arrangement includes an emotor having a rotor that is connected to the crankshaft and a stator that is fixed relative to the engine. A flex plate is connected for rotation with the rotor, and a resolver drive shaft extends from the flex plate, coaxial with the rotor. A resolver is provided having a resolver rotor that is connected with the resolver drive shaft at a position spaced apart from the flex plate, and a resolver stator is fixed relative to the emotor stator and located in proximity to the resolver rotor. A resolver rotor support acts to support the resolver drive shaft in proximity to the resolver stator. With this arrangement, a resolver rotor that is spaced apart from the crankshaft/emotor rotor connection rotates in a more stable manner, isolated from the deflections caused by the crankshaft so that the radial deflections of the resolver rotor are minimized in order to ensure proper functioning of the resolver and accurate readouts for the control functions of the emotor inverter.

[0006] In one embodiment, the flex plate includes a multi-plate stack.

[0007] In one embodiment, the resolver rotor support comprises a bearing or a bushing.

[0008] In one embodiment, an accessory element is drivingly connected to the resolver drive shaft. The accessory element can be a pump, and the resolver drive shaft extends through a pump housing of the pump. The pump can be used to deliver cooling fluid to the emotor stator. Alternatively, other types of accessory drive elements can be attached to the resolver drive shaft.

[0009] In one embodiment, the resolver rotor is located on an opposite side of the pump housing from the flex plate. In order to provide a space saving arrangement, the pump housing can be located at least partially within the rotor.

[0010] In one embodiment, a fixed support extends from a region of the stator, and the resolver stator is mounted to the fixed support. The resolver rotor support can be mounted on the fixed support.

[0011] In another aspect, an emotor assembly for a hybrid drive arrangement is provided. The emotor assembly includes a rotor that is adapted to be connected to a crankshaft and a stator that is fixed to an emotor housing. A flex plate is connected for rotation with the rotor, and a resolver drive shaft extends from the flex plate coaxial with the rotor. A resolver rotor is connected with the resolver drive shaft at a position spaced apart from the flex plate, and a resolver stator is fixed relative to the emotor housing and located in proximity to the resolver rotor. A resolver rotor support acts to support the resolver drive shaft in proximity to the resolver stator. The components of the emotor assembly can be preassembled as a unit during manufacture, and the thus pre-assembled emotor assembly can be installed as a single assembly onto an ICE of a hybrid vehicle drive system.

[0012] In one embodiment, the flex plate includes a multi-plate stack.

[0013] In one embodiment. The resolver rotor support comprises a bearing or a bushing.

[0014] In one embodiment, an accessory element is drivingly connected to the resolver drive shaft. The accessory element can be a pump, and the resolver drive shaft extends through a pump housing of the pump. The resolver rotor can be located on an opposite side of the pump housing from the flex plate, and the pump housing can be located at least partially within the rotor.

[0015] In one embodiment, a fixed support extends from the emotor housing in a region of the stator, and the resolver stator is mounted to the fixed support. The resolver rotor support can also be mounted on the fixed support.

[0016] Various features of the invention can be used alone or in combination in order to achieve one or more of the benefits described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The foregoing Summary and the following detailed description will be better understood when read in conjunction with the appended drawings, which illustrate preferred embodiments according to the disclosure. In the drawings:

[0018] FIG. 1 is a cross-sectional view of a hybrid drive arrangement for an internal combustion engine in accordance with a first embodiment of the disclosure.

[0019] FIG. 2 is a cross-sectional view of a hybrid drive arrangement for an internal combustion engine in accordance with a second embodiment of the disclosure.

[0020] FIG. 3 is a cross-sectional view of a hybrid drive arrangement for an internal combustion engine in which the resolver drive shaft also drives an accessory element.

DETAILED DESCRIPTION

[0021] Certain terminology is used in the following description for convenience only and is not limiting. The words “inwardly” and “outwardly” refer to directions toward and away from the parts referenced in the drawings. “Axially” refers to a direction along the axis of a shaft. “Radially” refers to a direction normal to an axis. A reference to a list of items that are cited as, for example, “at least one of a or b” (where a and b represent the items being listed) means any single one of the items a or b, or a combination of a and b thereof. This would also apply to lists of three or more items in like manner so that individual ones of the items or combinations thereof are included. The terms “about” and “approximately” encompass + or -10% of an indicated value unless otherwise noted. The terminology includes the words specifically noted above, derivatives thereof and words of similar import.

[0022] Referring to FIG. 1, a hybrid drive arrangement 20 for an internal combustion engine, indicated at 10 and includes a crankshaft 12, is shown. The hybrid drive arrangement 20 includes an emotor assembly 22 (referred to as emotor 22) having a housing 23 in which a rotor 24 is located that is connected to the crankshaft 12. This connection can be via connection to a bolt

flange **13** on the crankshaft **12** as shown or via other suitable means. The housing **23** can be made of one or more housing parts that area connected together. A stator **26** is fixed relative to the engine **10** (once the emotor housing **23** within which it is mounted is connected to the engine **10**), with the rotor **24** being mounted for rotation within the stator **26**.

[0023] A flex plate **30** is connected for rotation with the rotor **24**. As shown, the flex plate **30** can be connected to the rotor **24** via fasteners **32**, such as bolts or rivets that extend through the flex plate **30** as well as a rotor hub **25**. The flex plate **30** can be formed from spring steel, and can be a stamped part.

[0024] As shown in FIG. **1**, a resolver drive shaft **34** extends from the flex plate **30** coaxial with the rotor **24**. A resolver **40** is provided in order to track a position and angular velocity of the rotor **24** and signals speed and position information to an inverter that controls the emotor **22**. The resolver **40** includes a resolver rotor **42** that is connected with the resolver drive shaft **34** at a position spaced apart from the flex plate **30**. A resolver stator **44**, which is used to pick up position signals from the resolver rotor **42**, is fixed relative to the emotor stator **26** and located in proximity to the resolver rotor **42**. The resolver stator **44** can be mounted, for example, on a fixed support **54** that extends from a region of the stator **26**. This fixed support **54** can be part of the emotor housing **23** or fastened to the emotor housing **23**. A resolver rotor support **46** supports the resolver drive shaft **34** in proximity to the resolver stator **44**. In the illustrated embodiment, the resolver rotor support **46** is mounted on the fixed support **54**. However, it could be supported in another manner that is fixed relative to the engine **10** and/or the stator **26**.

[0025] The resolver rotor support **46** may include a bearing or bushing **48** in order to reduce friction as the resolver drive shaft **34** rotates with the rotor **24** of the emotor **22**.

[0026] With this arrangement, during use any deflections generated from the crankshaft mounting of the emotor rotor **24** are isolated from the resolver rotor **42** due to the flex plate **30** and the resolver rotor support **46**. This allows an air gap, typically in the 1-2 mm range, between the resolver rotor **42** and the resolver stator **44** to remain generally constant, improving the accuracy of the resolver and thee signals sent to the inverter for the motor **22**.

[0027] Referring now to FIG. **2**, a second embodiment of the hybrid drive arrangement **20'** is shown. The second embodiment of the hybrid drive arrangement **20'** is the same as the first embodiment of the hybrid drive arrangement **20** and like elements have been indicated with the same element numbers and the same description and function noted above also applies. The differences in the emotor **22'** are described below.

[0028] As shown in FIG. **2**, the flex plate **30'** is formed as a multi-plate stack including plates **31a'**, **31b'**, **31c'** that are fastened together in order to provide more flex in the flex plate **30'** than the single plate flex plate **30** in the first embodiment. This has the same function of allowing the resolver rotor **42** to rotate with a generally constant air gap between the resolver rotor **42** and the resolver stator **44**. This ensures accurate pickup of position and angular velocity of the resolver rotor **42** by the resolver stator **44** as the rotor **24** of the emotor **22'** is rotated. Isolating vibrations from the crankshaft **12** and/or other sources from the engine **10** enables more consistent functioning of the resolver **40** in determining the position of the rotor **24** and the angular velocity without inaccuracies and/or defaults based on varying air gaps in arrangements where the resolver rotor is mounted directly with the emotor rotor, particularly for cantilevered mounting arrangements as shown that are typical of P0 and P1 hybrid drive arrangements.

[0029] In the illustrated embodiment, three flex plates **31a'**, **31b'**, **31c'** are shown. Inner ends of the first flex plate **31a'** are attached to inner ends of the second flex plate **31b'**, and outer ends of the second flex plate **31b'** are connected to outer ends of the third flex plate **31c'** to which the resolver drive shaft **34** is mounted. Here, outer ends of the first flex plate **31a'** are connected to the rotor **24** of the emotor **22'**. As will be recognized by those skilled in the art, the number of flex plates as well as their spring form/spring factor can be adjusted depending upon the particular application.

[0030] Referring now to FIG. **3**, a third embodiment of a hybrid drive arrangement **20''** is shown.

The third embodiment of the hybrid drive arrangement **20''** is substantially the same as the second embodiment of the hybrid drive arrangement **20'** and like elements are indicated with the same reference numerals and the same description and function noted above also applies. The differences in the emotor **22''** are described below.

[0031] As shown in FIG. 3, an accessory element **50''** is drivingly connected to the resolver drive shaft **34** and is driven thereby. The accessory element **50''** may be an oil pump that is used to spray cooling oil on the stator **26**. However, it could also be a different accessory drive element.

[0032] In the illustrated embodiment, the resolver drive shaft **34** extends through a pump housing **52''** of the accessory element **50''** which is a pump. Here, the resolver rotor **42** is located on an opposite side of the pump housing **52''** from the flex plate **30'**. In order to save space, the pump housing **52''** can be located at least partially within the rotor **24**. In the illustrated embodiment, the resolver rotor support **46''** is in the form of a bushing or bearing **48''** that is located within the housing **52''** of the pump **50''**.

[0033] For each of the embodiments of the hybrid drive arrangement **20**, **20'**, **20''**, the emotor assembly **22**, **22'**, **22''** can be pre-assembled as a unit that is then connected to the internal combustion engine **10**. This saves assembly time during final assembly of the internal combustion engine **10**. Here, the emotor assembly **22**, **22'**, **22''** includes the rotor **24** which is adapted to be connected to the crankshaft **12** as well as the stator **26** that is affixed to the emotor housing **23**. The flex plate **30**, **30'** (shown in FIGS. 1 and 2) is connected for rotation with the rotor **24** and the resolver drive shaft **34** extends from the flex plate **30**, **30'** co-axial with the rotor **24**. The resolver **40** is assembled in the emotor assembly **22**, **22'**, **22''** with the resolver rotor **42** connected to the resolver drive shaft **30** at a position spaced part from the flex plate **30**, **30'**. The resolver stator **44** is fixed relative to the emotor housing **23**, for example with the fixed support **54** that can be a part of or attached to the emotor housing **23**. The resolver rotor support **46**, **46''** supports the resolver drive shaft **34** in proximity to the resolver stator **44** in order to avoid radial shifting of the resolver rotor **42** which can cause an inconsistent air gap between the resolver rotor **42** and the resolver stator **44** that can cause issues in connection with the transformation ratio leading to signal processing issues and speed read out issues as discussed above in the inverter controlling the emotor **22**, **22'**, **22''**. To the extent provided, the pump **50''** can also be preassembled in the emotor assembly **22''**

[0034] Having thus described the presently preferred embodiments in detail, it is to be appreciated and will be apparent to those skilled in the art that many physical changes, only a few of which are exemplified in the detailed description, could be made without altering the inventive concepts and principles embodied therein. It is also to be appreciated that numerous embodiments incorporating only part of the preferred embodiment are possible which do not alter, with respect to those parts, the inventive concepts and principles embodied therein. The present embodiments and optional configurations are therefore to be considered in all respects as exemplary and/or illustrative and not restrictive, the scope that is indicated by the appended claims rather than by the foregoing description, and all alternate embodiments and changes to this embodiment which come within the meaning and range of equivalency of said claims are therefore to be embraced therein.

LIST OF REFERENCE SYMBOLS

[0035] **10** internal combustion engine [0036] **12** crankshaft [0037] **20**, **20'**, **20''** hybrid drive arrangement [0038] **22**, **22'**, **22''** emotor [0039] **23** emotor housing [0040] **24** rotor [0041] **25** rotor hub [0042] **26** stator [0043] **30**, **30'** flex plate [0044] **31a'**, **31b'**, **31c'** plates [0045] **32** fasteners [0046] **34** resolver drive shaft [0047] **40** resolver [0048] **42** resolver rotor [0049] **44** resolver stator [0050] **46**, **46''** resolver rotor support [0051] **48**, **48''** bearing or bushing [0052] **50''** accessory element/pump [0053] **52''** pump housing [0054] **54** fixed support

Claims

- 1.** A hybrid drive arrangement for an internal combustion engine having a crankshaft, the hybrid drive arrangement comprising: an emotor having a rotor that is connected to the crankshaft and a stator that is fixed relative to the engine; a flex plate connected for rotation with the rotor; a resolver drive shaft extending from the flex plate coaxial with the rotor; a resolver rotor connected with the resolver drive shaft at a position spaced apart from the flex plate; a resolver stator fixed relative to the emotor stator and located in proximity to the resolver rotor; and a resolver rotor support that supports the resolver drive shaft in proximity to the resolver stator.
 - 2.** The hybrid drive arrangement of claim 1, wherein the flex plate includes a multi-plate stack.
 - 3.** The hybrid drive arrangement of claim 1, wherein the resolver rotor support comprises a bearing or a bushing.
 - 4.** The hybrid drive arrangement of claim 1, further comprising an accessory element drivingly connected to the resolver drive shaft.
 - 5.** The hybrid drive arrangement of claim 4, wherein the accessory element is a pump, and the resolver drive shaft extends through a pump housing of the pump.
 - 6.** The hybrid drive arrangement of claim 5, wherein the resolver rotor is located on an opposite side of the pump housing from the flex plate.
 - 7.** The hybrid drive arrangement of claim 5, wherein the pump housing is located at least partially within the rotor.
 - 8.** The hybrid drive arrangement of claim 1, further comprising a fixed support extending from a region of the stator, and the resolver stator is mounted to the fixed support.
 - 9.** The hybrid drive arrangement of claim 1, wherein the resolver rotor support is mounted on the fixed support.
 - 10.** An emotor assembly for a hybrid drive arrangement, the emotor assembly comprising: a rotor that is adapted to be connected to a crankshaft; a stator that is fixed to an emotor housing; a flex plate connected for rotation with the rotor; a resolver drive shaft extending from the flex plate coaxial with the rotor; a resolver rotor connected with the resolver drive shaft at a position spaced apart from the flex plate; a resolver stator fixed relative to the emotor housing and located in proximity to the resolver rotor; and a resolver rotor support that supports the resolver drive shaft in proximity to the resolver stator.
 - 11.** The emotor assembly of claim 10, wherein the flex plate includes a multi-plate stack.
 - 12.** The emotor assembly of claim 10, wherein the resolver rotor support comprises a bearing or a bushing.
 - 13.** The emotor assembly of claim 10, further comprising an accessory element drivingly connected to the resolver drive shaft.
 - 14.** The emotor assembly of claim 13, wherein the accessory element is a pump, and the resolver drive shaft extends through a pump housing of the pump.
 - 15.** The emotor assembly of claim 14, wherein the resolver rotor is located on an opposite side of the pump housing from the flex plate.
 - 16.** The emotor assembly of claim 14, wherein the pump housing is located at least partially within the rotor.
 - 17.** The emotor assembly of claim 10, further comprising a fixed support extending from a region of the stator, and the resolver stator is mounted to the fixed support.
 - 18.** The emotor assembly of claim 10, wherein the resolver rotor support is mounted on the fixed support.
-